D02 · Capacity & Long-Range Planning

04 · Make-vs-buy & co-manufacturing strategy

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1 WHAT IT IS

The structural decision of which capacity the organization owns and which it rents from third parties — co-manufacturers, contract manufacturers, EMS providers, OSAT partners. It is not a unit-cost comparison but a long-term commitment between vertical integration (own the asset, own the people, own the IP) and contracted capacity (own the recipe, rent the asset, share the IP boundary). The decision compounds: every product family treated this way creates dependencies that take years to reverse.

2 WHY IT MATTERS

Make-vs-buy is the decision most often reduced to a spreadsheet that should not be. **Transaction cost economics treats the choice as a function of asset specificity, contracting frequency, and information asymmetry (Williamson, 1985 — Nobel-recognized framework)** — categories that do not appear in a unit-cost comparison but determine whether the rented capacity will still be there in three years. The dominant failure mode is structural: buying when the asset is specific to the company's IP (you lock yourself into a supplier who learns your know-how) and making when the asset is generic (you commit capex to a commodity you could have rented). The other axis is volatility — **own the capacity that absorbs the base load, rent the capacity that absorbs the peaks** — because depreciation is a fixed cost and demand variability is not.

3 HOW IT IS DONE

1. **Classify each product family by asset specificity and IP sensitivity.** Highly specific assets (custom tooling, proprietary processes) and high-IP-content products belong in-house; generic assets and commodity processes are candidates to outsource. This is the screen before the financial case.
2. **Split base load from peak demand for each family.** The case for owning capacity is strongest where utilization is stable; the case for renting is strongest where demand is volatile. The hybrid model — own enough capacity for the base, rent the peaks — is structurally superior in most consumer-driven categories.
3. **Compute total cost of ownership, not unit cost.** Internal manufacturing carries fixed depreciation, plant overhead, supervision, quality systems. Co-manufacturing carries unit cost premium, technology transfer cost, qualification cycles, ongoing audit and management. The unit-cost comparison that ignores these categories systematically favors outsourcing.
4. **Stress-test the contract under disruption scenarios.** What happens to your co-manufacturer's capacity allocation when their largest customer doubles its demand? What is your priority in their queue when raw material is constrained? Pricing flexibility, capacity reservation contracts, and exit clauses determine whether the relationship survives stress.
5. **Build the dual-source posture deliberately, not by accident.** A single co-manufacturer is a single point of failure. The cost of qualifying a second source is not optional — it is the insurance premium against losing the first one. Plan for 70/30 or 80/20 volume allocation, not 100/0.
6. **Review the portfolio annually with an explicit re-shoring lens.** The conditions that justified the original make-vs-buy decision change — labor cost convergence, tariff regimes, technology automation, geopolitical risk. The annual review is where decisions made in 2018 get re-tested against 2026 reality.

IN PRACTICE

Run the TCO with the hidden costs

Build the comparison on total cost of ownership — including the coordination, quality-oversight, IP-leakage, and switching costs the unit-price comparison omits. The buy option that looks cheaper per unit often carries transaction costs that reverse the conclusion once they are made visible.

Dual-source what you cannot survive losing

For bought capacity that is critical and hard to replace, maintain a qualified second source. Single-sourcing critical rented capacity recreates the lock-in the buy decision was supposed to avoid; the dual source is the price of keeping the option to leave.

Re-evaluate the portfolio annually

Revisit the make-vs-buy portfolio each year against changing cost, tariff, and geopolitical realities. The boundary of the firm is not set once — a make decision that was right under one cost and tariff regime can become wrong under another, and nearshoring shifts the calculus on which way the boundary should move.

Govern co-manufacturing as managed capacity

Treat co-manufacturers as part of the capacity plan, with capacity commitments, quality governance, and forward visibility — the same rigor applied to internal capacity and supplier commitment. Rented capacity managed loosely behaves like an uncontrolled variable; managed tightly it is a genuine flexibility asset.

WHAT YOU WOULD USE

MAKE-VS-BUY / SOURCING STRATEGY

o9 / SAP IBP / Kinaxis — network & sourcing modeling — Capacity and sourcing scenario modeling that compares make, buy, and co-manufacturing options inside the integrated plan with cost and capacity impact. **(o9, SAP, Kinaxis — vendor)** The choice when the make-vs-buy portfolio must reconcile with the capacity and supply plan.

Anaplan / cost-modeling platforms — Flexible TCO and scenario-modeling platforms for sourcing and footprint decisions. **(Anaplan, cost-modeling vendors — vendor)** A strong fit where the decision is driven by financial TCO and scenario comparison across options.

Should-cost / sourcing-analytics specialists — Dedicated should-cost and sourcing-analytics tools that quantify the true cost of buy options. **(specialized vendors — vendor)** The vertical option when rigorous should-cost modeling of suppliers is the deciding input.

For operations that don't need a platform — A TCO comparison in a workbook that scores each capability on asset specificity and volatility, with the hidden transaction costs made explicit. The framework — own the specific and stable, rent the generic and volatile — is the decision; the tool only does the arithmetic the framework defines.

4 SO WHAT

Make-vs-buy is the structural decision that determines what kind of company you are. Own what is specific to your IP and stable in volume; rent what is generic and volatile. Run the TCO with all the costs the unit-price comparison hides, dual-source what you cannot survive losing, and re-evaluate the portfolio annually against changing cost, tariff, and geopolitical realities. The boundary of the firm is a strategic choice, not an accident of history.

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